

ECON 101: Macroeconomics

Practice Questions and Answers

Question: Name three products whose demand is likely to increase rapidly if the following demographic groups increase at a faster rate than the population as a whole:

- a. Teenagers
- b. Children under age five
- c. People over age sixty-five

Answer:

- a. Music downloads, video streaming, text messaging services, video game subscriptions
- b. Diapers, developmental toys, child care services
- c. Medical services, cruise ship tours, nursing homes

Demand and Supply Equations

The first step in a quantitative analysis is to supplement our use of demand and supply curve with demand and supply equations. For example, suppose that economists have estimated that the demand for apartments in Toronto is: $Q^D = 3000000 - 1000P$ and the supply of apartments is: $Q^S = -450000 + 1300P$.

We have used Q^D for the quantity of apartments demanded per month, Q^S for the quantity of apartments supplied per month, and P for the apartment rent, in dollars per month. The competitive market equilibrium occurs when the quantity demanded equals the quantity supplied, or: $Q^D = Q^S$.

We can use this equation, which is called an *equilibrium condition*, to solve for the equilibrium monthly apartment rent by setting the quantity demanded from the demand equation equal to the quantity demanded from the supply equation:

$$3000000 - 1000P = -450000 + 1300P$$

$$P = 3450000/2300 = \$1500.$$

We can then substitute this price back into either the demand equation or the supply equation to find the equilibrium quantity of apartments rented:

$$Q^D = 3000000 - 1000P = 3000000 - 1000(1500) = 1500000$$

$$Q^S = -450000 + 1300P = -450000 + 1300(1500) = 1500000.$$

Table 1

Price per Bushel	Quantity Demanded (bushels)	Quantity Supplied (bushels)
\$3	30,000	0
6	26,000	4,000
9	22,000	9,000
12	18,000	12,000
15	15,000	15,000
18	12,000	22,000
21	8,000	28,000
24	4,000	36,000

27) **Refer to Table 1.** The table contains information about the corn market. Use the table to answer the following questions.

- What are the equilibrium price and quantity of corn?
- Suppose the prevailing price is \$9 per bushel. Is there a shortage or a surplus in the market?
- What is the quantity of the shortage or surplus?
- How many bushels will be sold if the market price is \$9 per bushel?
- If the market price is \$9 per bushel, what must happen to restore equilibrium in the market?
- At what price will suppliers be able to sell 22,000 bushels of corn?
- Suppose the market price is \$21 per bushel. Is there a shortage or a surplus in the market?
- What is the quantity of the shortage or surplus?
- How many bushels will be sold if the market price is \$21 per bushel?
- If the market price is \$21 per bushel, what must happen to restore equilibrium in the market?

Answer:

- Equilibrium price = \$15; Equilibrium quantity = 15,000 bushels.
- There is a shortage.
- Shortage = $22,000 - 9,000 = 13,000$ bushels.
- Quantity sold = 9,000 bushels.
- Price must rise.
- At \$9 per bushel
- There is a surplus.
- Surplus = $28,000 - 8,000 = 20,000$ bushels.
- Quantity sold = 8,000 bushels.
- Price must fall.